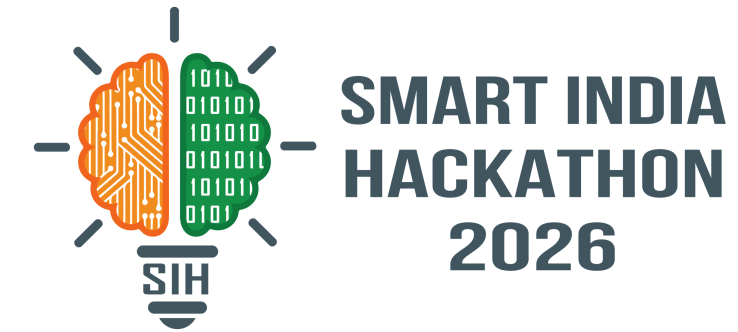


SMART INDIA HACKATHON 2026



BLACK BOX

- **Problem Statement ID – SIH26220**
- **Problem Statement Title - student innovation-creating intelligent devices to improve commutation sector**
- **Theme- Smart Vehicles**
- **PS Category- Hardware**
- **Team ID-**
- **Team Name- Black Box**



INTELLIGENT ACCIDENT AND ROAD HAZARD MONITORING SYSTEM

PROPOSED SOLUTION:

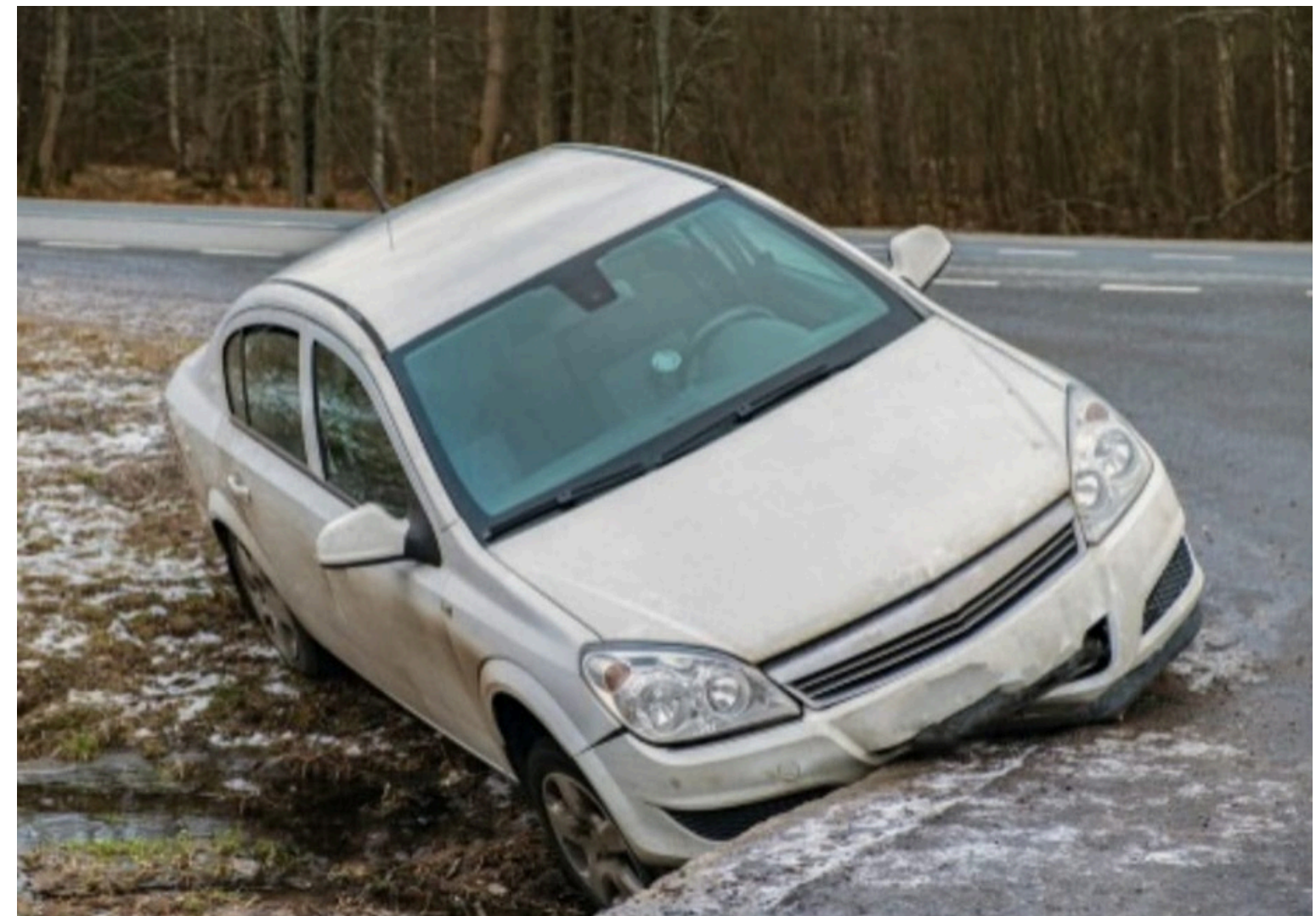
- Detect accidents & road hazards.
- Record motion, speed, time & GPS.
- Store data + trigger alerts.

PROBLEM SOLVED:

- Accident evidence + hazard mapping.
- Faster location-based response.

UNIQUE INNOVATION:

- One device: Crash Recorder + Road Sensor.
- “Not just a crash recorder — a continuously learning road-safety sensor.”

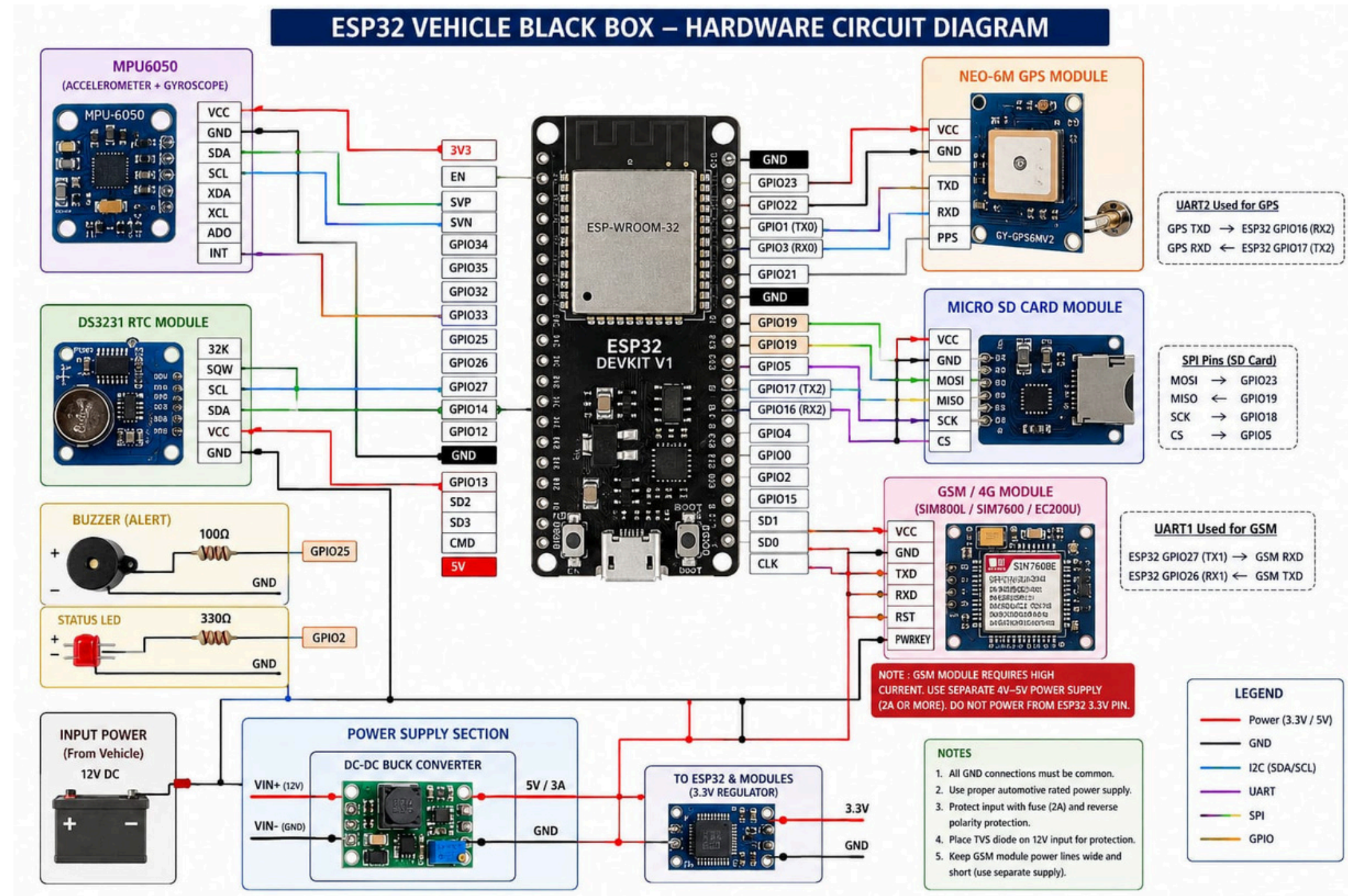


TECHNOLOGIES:

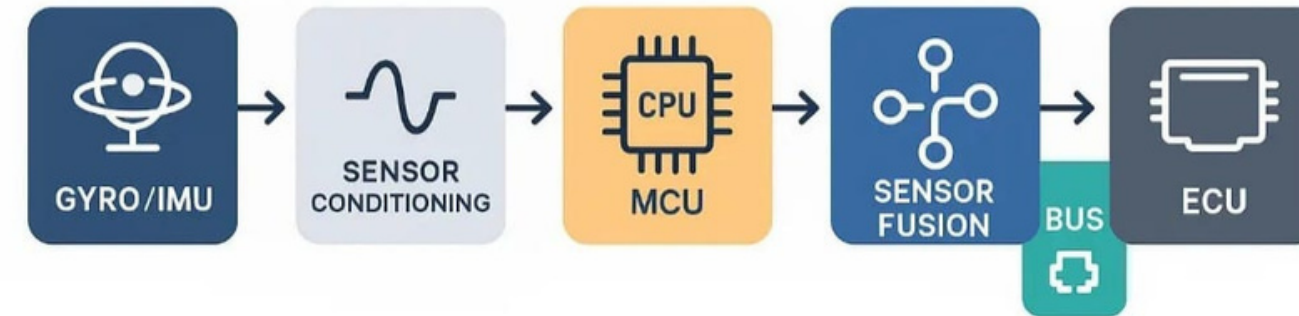
- ESP32 — Edge controller
- MPU6050 — Motion & impact sensing
- NEO-6M GPS — Location & speed
- MicroSD + DS3231 — Event storage & timestamp
- GSM/4G — Emergency communication

Software:

- Embedded C/C++ • Arduino IDE •
- ML-based event classification



FEASIBILITY AND VIABILITY



FEASIBILITY:

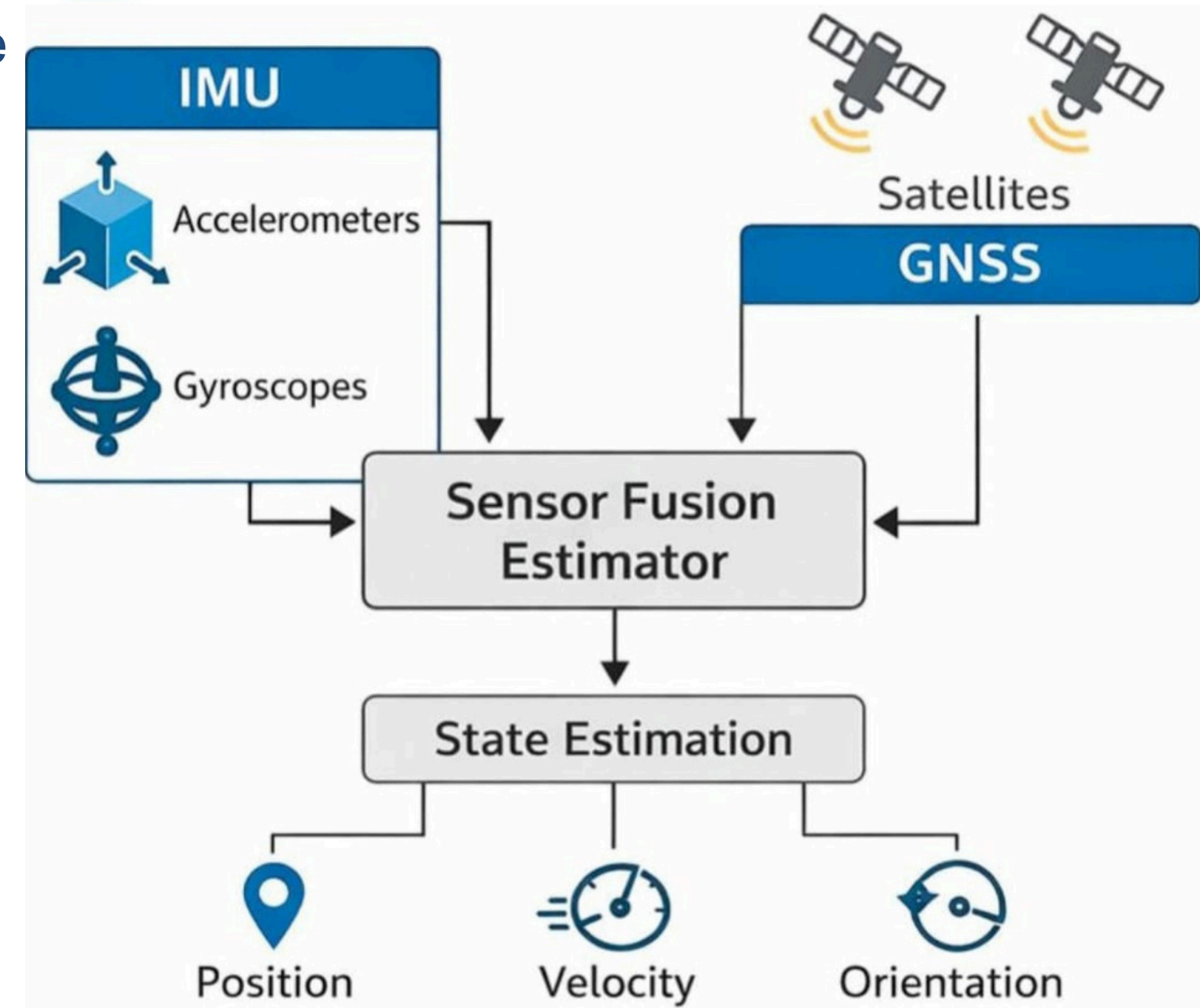
- Prototype-ready using low-cost, readily available hardware
- Compact, modular & easy to retrofit
- Uses proven embedded technologies

POTENTIAL CHALLENGES & RISKS:

- False detection from bumps & hard braking
- GPS/GSM signal loss
- Vehicle power fluctuations

STRATEGIES TO OVERCOME:

- False alerts → Sensor fusion + calibration
- Network loss → Local SD data logging
- Power issues → Regulated supply + protection



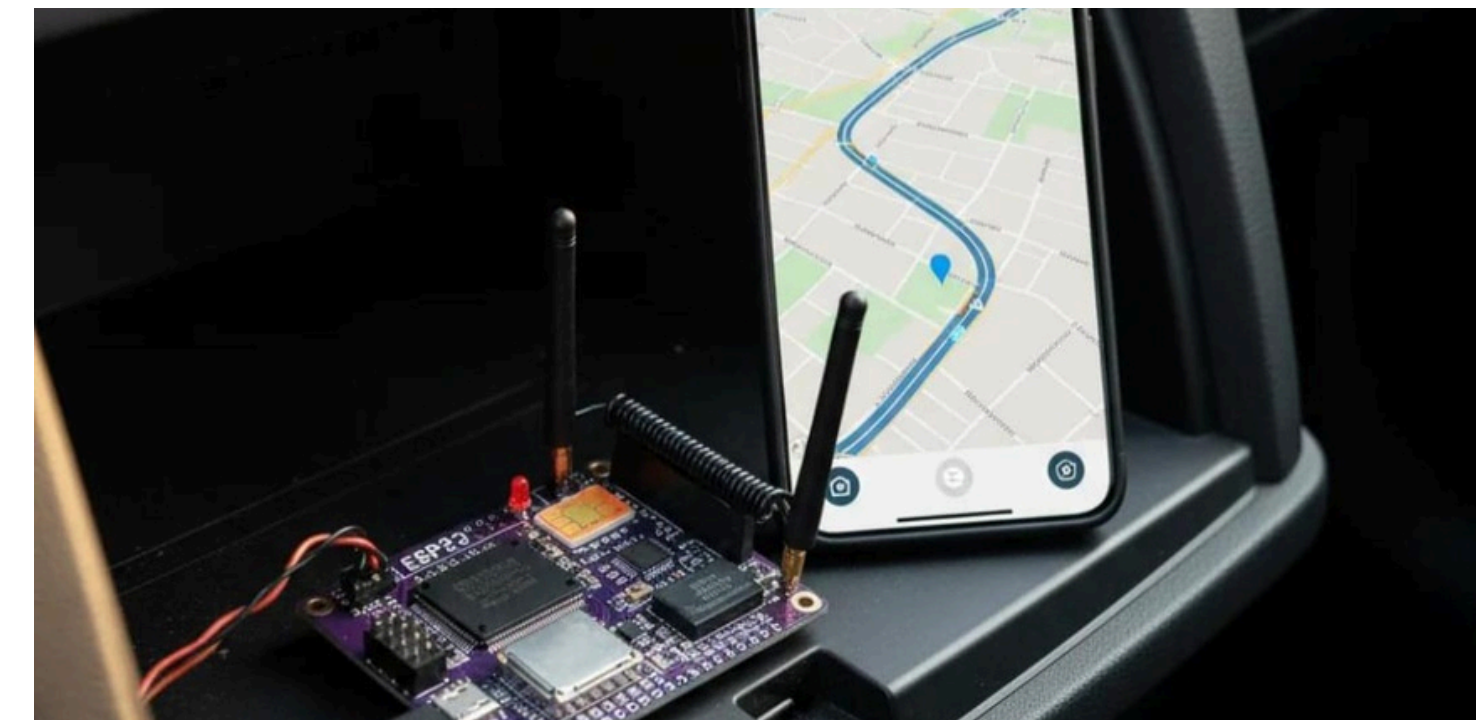
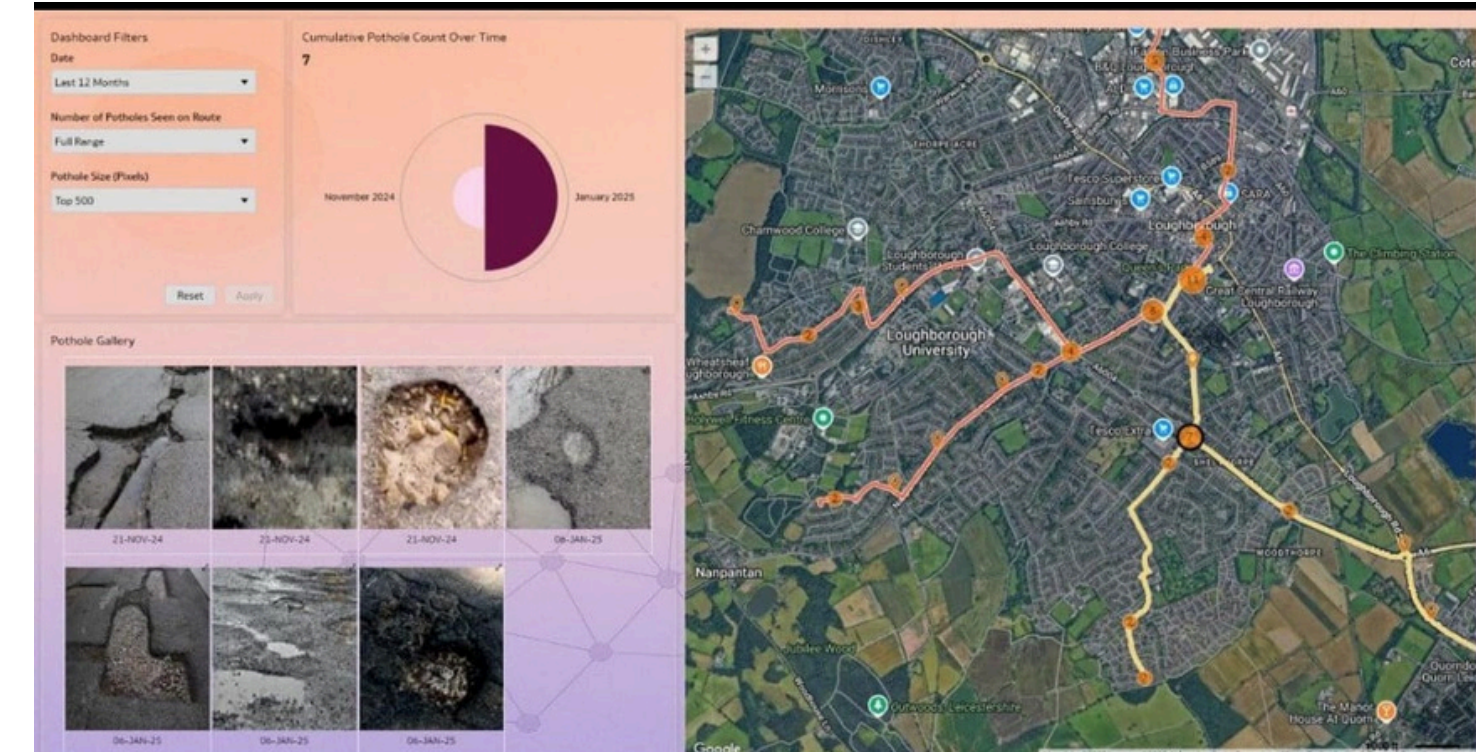
POTENTIAL IMPACT:

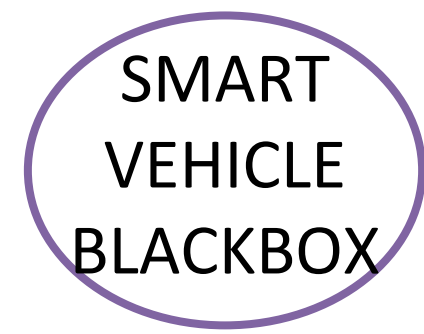
- Faster accident response through precise location alerts
- Safer roads through crowd-sourced hazard detection
- Reliable crash data for drivers & authorities

BENEFITS:

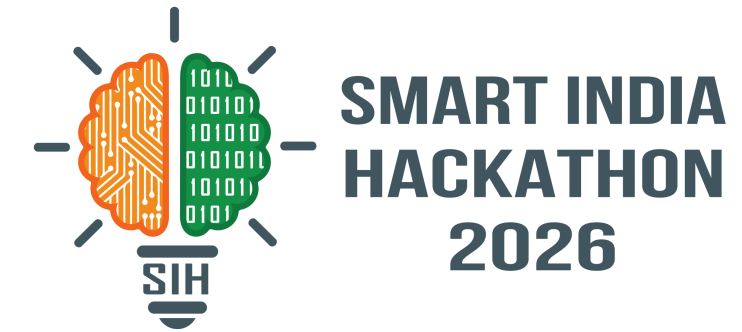
- Social: Improved road safety & emergency response
- Economic: Low-cost retrofit; reduces accident-related losses
- Environmental: Identifies road damage for targeted maintenance

“Every journey generates data that can make the next journey safer.”





RESEARCH AND REFERENCES



NHTSA — Event Data Recorders (EDR),
Crash data recording, vehicle dynamics & post-crash information.

NHTSA — Event Data Recorder

Sensors — Automated ML Pothole Detection
Uses vibration/accelerometer + GPS + machine learning for pothole detection and geo-location.

ScienceDirect — Road Surface Assessment
Uses GPS + accelerometer to identify and locate bumps/potholes.

